

Early Prediction of Prolonged ICU Length of Stay Using MIMIC-IV Demo

1. Introduction

Efficient ICU resource management is one of the most important operational challenges faced by modern hospitals. Patients who remain in the Intensive Care Unit (ICU) for prolonged periods consume a disproportionate amount of medical resources, including hospital beds, nursing care, medications, and specialized equipment. Being able to identify these patients early may help clinicians improve discharge planning, optimize resource allocation, and prioritize high-risk patients for additional monitoring.

The objective of this project is to develop an end-to-end machine learning pipeline that predicts whether an ICU stay will become prolonged using only information available during the first 24 hours after ICU admission. Rather than focusing solely on predictive performance, this project emphasizes building a reproducible, interpretable, and leakage-free data science workflow that closely resembles real-world healthcare analytics practice.

Because access to the full MIMIC-IV database requires credentialing and is governed by a Data Use Agreement, this project uses the publicly available **MIMIC-IV Clinical Database Demo v2.2**. Although the demo dataset contains only approximately 100 patients, it preserves the original relational database structure and provides an excellent environment for demonstrating SQL-based feature engineering, multi-table data integration, clinical feature extraction, and machine learning pipeline design.

2. Project Objectives

The primary objective of this project is not to maximize model accuracy on a small demonstration dataset. Instead, the project focuses on demonstrating the complete lifecycle of a clinical machine learning project.

The project aims to:

- Build a leakage-free cohort using ICU admission data.
- Integrate multiple relational tables into a patient-level analytical dataset.
- Design clinically meaningful features that are available within the first 24 hours of ICU admission.
- Implement a reproducible SQL-based feature extraction pipeline.
- Prepare a high-quality modeling dataset suitable for downstream machine learning.
- Produce an interpretable prediction model that can later be explained using SHAP.

3. Dataset

This project uses the **MIMIC-IV Clinical Database Demo v2.2**, released by the MIT Laboratory for Computational Physiology through PhysioNet.

The demo dataset is an openly accessible subset of the full MIMIC-IV database and contains approximately 100 patients and 140 ICU stays. Although much smaller than the full database, it preserves the same schema and relational structure, making it well suited for demonstrating data engineering and machine learning workflows.

Six tables were selected for this project.

Table	Purpose
patients	Patient demographics
admissions	Hospital admission information
icustays	Defines ICU stays and ICU length of stay
diagnoses_icd	Chronic disease identification
labevents	Laboratory measurements
prescriptions	Medication records

Several other MIMIC-IV tables, including chart events, input/output events, microbiology events, and billing-related tables, were intentionally excluded from the initial version of the project because they either introduce unnecessary complexity, substantially increase computational cost, or contain information unavailable during the first 24 hours of ICU admission.

4. Prediction Target

The prediction task is formulated as a binary classification problem.

Each row in the final dataset represents one ICU stay.

The target variable is:

prolonged_stay_label

where

- 1 = ICU Length of Stay ≥ 5 days

- 0 = ICU Length of Stay < 5 days

The five-day threshold was selected after exploratory profiling of the ICU length-of-stay distribution. Approximately the 75th percentile of ICU stays occurred around five days, resulting in a reasonably balanced classification problem while maintaining clinical relevance.

To ensure that the prediction mimics a real clinical decision-making scenario, only information available within the first 24 hours after ICU admission is used as model input. Any information recorded after this time window is excluded to prevent data leakage.

5. Cohort Construction

The unit of analysis in this project is an ICU stay rather than an individual patient.

The cohort construction process consists of two major steps.

First, ICU stays lasting less than one day were excluded because these encounters often represent brief observation stays, transfers, or administrative admissions that are unlikely to benefit from prolonged-stay prediction.

After cohort selection, the dataset contained:

- 117 ICU stays
- 117 unique stay_id values

Each stay was assigned a binary prolonged-stay label based on the predefined five-day threshold.

6. SQL-Based Feature Engineering Pipeline

One of the primary goals of this project is to demonstrate practical SQL skills when working with complex relational healthcare databases.

Instead of extracting features directly within Python, feature generation was organized into independent SQL modules. Each feature domain was implemented separately and later merged into a single analytical dataset.

The feature extraction pipeline consists of five domains:

- Demographic features
- Admission context features
- Comorbidity features
- Laboratory features (first 24 hours)
- Medication features (first 24 hours)

All laboratory and medication features were explicitly restricted to records occurring within the first 24 hours following ICU admission. This temporal restriction prevents future information from entering the prediction pipeline and ensures that the resulting dataset is suitable for prospective prediction.

The completed SQL pipeline generated a final feature table containing:

- 117 ICU stays
- 17 predictive features
- 19 total columns (including stay_id and target)

Automatic validation was performed after feature extraction. Validation confirmed that all stay identifiers were unique, no duplicate rows were introduced during joins, and the final row count exactly matched the study cohort.

7. Feature Overview

The final analytical dataset contains five categories of predictive features.

Demographic

- Age
- Gender

Admission Context

- Admission type
- Insurance
- Night admission
- Weekend admission

Comorbidities

- Diabetes
- Hypertension
- Chronic kidney disease
- Heart failure
- Comorbidity count

Laboratory Tests

- First creatinine
- First white blood cell count
- First hemoglobin

Medication Features

- Medication count
- Antibiotic flag
- Insulin flag

The next stage of the project focuses on exploratory data analysis (EDA), feature preprocessing, model development, and model interpretation.

8. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted on the final analytical dataset after completion of the SQL feature extraction pipeline. The primary objective of this stage was to evaluate data quality, understand feature distributions, examine relationships between predictors and the target variable, and identify preprocessing requirements before machine learning modeling.

8.1 Dataset Overview

The final analytical dataset contained **117 ICU stays** and **17 predictive features**, together with one identifier (**stay_id**) and one binary target variable (**prolonged_stay_label**), resulting in a total of **19 columns**.

The feature set consisted of five domains:

- Demographic characteristics
- Admission context
- Chronic comorbidities
- Laboratory measurements obtained within the first 24 hours
- Medication-related features obtained within the first 24 hours

The target variable was moderately imbalanced, with **31 prolonged ICU stays (26.5%)** and **86 non-prolonged ICU stays (73.5%)**, making the dataset appropriate for stratified model evaluation.

8.2 Data Quality Assessment

Data quality checks confirmed that the analytical dataset was suitable for downstream modeling.

Validation showed that:

- No duplicated ICU stay identifiers were present.
- No duplicated observations were generated during SQL joins.
- No missing values remained after feature extraction.
- All categorical and binary variables contained valid values.

Unlike many clinical datasets, missing laboratory measurements had already been addressed during feature extraction. Consequently, no additional imputation was required during the EDA stage.

8.3 Feature Distributions

Numerical variables—including age, creatinine, white blood cell count, hemoglobin, medication count, and comorbidity count—were examined using descriptive statistics, histograms, and boxplots.

Several laboratory variables exhibited right-skewed distributions and contained extreme values. However, these observations were considered clinically plausible because critically ill ICU patients often present with severe physiological abnormalities. Therefore, these values were retained rather than treated as measurement errors.

Categorical variables showed reasonable distributions across demographic characteristics, admission types, insurance categories, and chronic disease indicators. Medication-related variables indicated that antibiotic exposure was common during the first 24 hours of ICU admission, reflecting the clinical characteristics of critically ill patients.

8.4 Relationship Between Features and the Prediction Target

Relationships between individual predictors and prolonged ICU stay were explored using boxplots for numerical variables and grouped bar charts for categorical variables.

Overall, most predictors demonstrated relatively weak univariate relationships with the outcome. This finding is expected in clinical prediction problems, where patient outcomes are typically influenced by multiple interacting factors rather than a single dominant variable.

Some clinically meaningful trends were observed. Patients with prolonged ICU stays generally exhibited higher creatinine levels, greater medication burden, and a higher prevalence of chronic comorbidities. Nevertheless, these differences should be interpreted descriptively rather than inferentially due to the limited sample size of the demo dataset.

8.5 Correlation Analysis

A correlation matrix was generated to evaluate relationships among numerical variables.

Overall, pairwise correlations were modest, suggesting limited multicollinearity within the numerical feature set. Expected clinical relationships were observed, such as moderate associations between comorbidity burden and chronic disease indicators.

Given the use of tree-based machine learning models in later stages of the project, mild correlations were not considered problematic.

8.6 Modeling Readiness

Based on the EDA results, the dataset was considered suitable for machine learning modeling.

The following preprocessing steps were identified for the next stage:

- One-hot encoding of categorical variables.
- Feature scaling for algorithms sensitive to variable magnitude (e.g., Logistic Regression).
- Preservation of binary clinical indicators without transformation.
- Stratified sampling and cross-validation to address the moderate class imbalance.

No additional data cleaning was deemed necessary before model development.

Overall, the EDA confirmed that the SQL feature engineering pipeline produced a high-quality analytical dataset ready for feature engineering and predictive modeling.

9. Feature Engineering and Preprocessing

Following exploratory data analysis, the frozen feature table was transformed into a model-ready dataset while preserving the temporal and leakage constraints established during the SQL feature extraction stage.

The final analytical dataset contained 117 ICU stays and 17 original predictive features. The target variable, `prolonged_stay_label`, represented whether an ICU stay lasted at least five days. Of the 117 stays, 31 (26.5%) were classified as prolonged stays and 86 (73.5%) as non-prolonged stays.

9.1 Feature Types and Encoding

The predictors were divided into numerical, categorical, and binary variables.

Five numerical variables were retained:

- `age_at_admission`
- `first_creatinine`
- `first_wbc`
- `first_hemoglobin`
- `medication_count`

Three categorical variables were encoded using one-hot encoding:

- `gender`
- `admission type`
- `insurance`

`drop_first=True` was used to avoid redundant dummy variables for the linear Logistic Regression model.

Eight binary clinical indicators, including diabetes, hypertension, heart failure, CKD, antibiotic exposure, and insulin exposure, were retained as 0/1 variables without further transformation.

The resulting model-ready dataset contained 117 rows and 23 columns, including the identifier, target variable, and encoded predictors.

9.2 Feature Scaling

Feature scaling was handled differently depending on the model family.

For Logistic Regression, numerical variables were standardized using a `ColumnTransformer` inside a scikit-learn Pipeline. Importantly, the scaler was fitted only on the training data rather than on the complete dataset. This prevented information from the test set from influencing the preprocessing stage.

Random Forest and XGBoost were not subjected to numerical scaling because tree-based models are generally insensitive to differences in feature magnitude.

The final `model_ready.csv` therefore remained unscaled, while model-specific preprocessing was performed inside the training pipeline. This design allows the same dataset to be used consistently across different model families without introducing preprocessing leakage.

9.3 Multicollinearity and Comorbidity Representation

A potential redundancy existed between the four individual comorbidity indicators and the derived `comorbidity_count` variable.

Variance Inflation Factor analysis showed that all features remained below 2.0 under both representations, indicating no severe multicollinearity. Nevertheless, the four individual comorbidity indicators were retained while `comorbidity_count` was removed.

This decision was made for two reasons. First, the individual indicators preserve more information than their aggregate count. Second, retaining individual conditions allows the model interpretation stage to identify which specific comorbidities contribute to predictions.

Therefore, the final feature representation prioritizes interpretability and clinical specificity rather than reducing the number of variables.

10. Machine Learning Modeling

10.1 Modeling Strategy

Three supervised classification models were evaluated:

1. Logistic Regression
2. Random Forest
3. XGBoost

Logistic Regression was selected as an interpretable linear baseline. Random Forest and XGBoost were included to capture potential nonlinear relationships and interactions that may not be represented by a linear decision boundary.

Because the dataset contained only 117 ICU stays and 31 positive cases, extensive hyperparameter tuning was deliberately avoided. Performing extensive optimization on such a small dataset could cause the model to overfit cross-validation noise rather than identify genuinely generalizable patterns.

The dataset was divided using a stratified train/test split, producing 93 training observations and 24 test observations. Stratification preserved the approximate proportion of prolonged and non-prolonged stays in both subsets.

In addition, five-fold stratified cross-validation was performed on the training set. Cross-validation was treated as the primary basis for model comparison because performance from a single 24-patient test set is highly sensitive to individual observations.

10.2 Test Set Performance

The test-set results were:

Model	ROC-AUC	Precision	Recall	F1
Logistic Regression	0.593	0.333	0.667	0.444
Random Forest	0.593	0.333	0.333	0.333
XGBoost	0.481	0.222	0.333	0.267

All three models produced ROC-AUC values close to the 0.50 no-skill reference level. Logistic Regression and Random Forest achieved the highest test-set ROC-AUC at 0.593, while XGBoost performed below the no-skill level with an ROC-AUC of 0.481.

These results should not be interpreted as evidence that the modeling pipeline failed. Rather, they are consistent with the weak feature-target relationships identified during EDA and the very small sample size of the MIMIC-IV Demo dataset.

The test set contained only 24 observations, including six prolonged stays. Consequently, a small number of classification changes can substantially affect precision, recall, and F1-score.

10.3 Cross-Validation Results

Five-fold stratified cross-validation produced the following results:

Model	CV ROC-AUC	CV F1
Random Forest	0.537 ± 0.083	0.337 ± 0.137
XGBoost	0.533 ± 0.082	0.272 ± 0.076
Logistic Regression	0.441 ± 0.141	0.326 ± 0.180

The cross-validation ranking differed slightly from the single test-set ranking. Random Forest achieved the highest mean ROC-AUC at 0.537, narrowly followed by XGBoost at 0.533. Logistic Regression had a lower mean ROC-AUC of 0.441 and substantially larger variability across folds.

This discrepancy illustrates the instability expected when cross-validation folds contain only approximately 19 observations each. Therefore, the small difference between Random Forest and XGBoost should not be interpreted as a statistically meaningful performance advantage.

10.4 Model Selection

Random Forest was selected as the final model based on its highest mean five-fold cross-validation ROC-AUC.

However, the difference between Random Forest and XGBoost was only approximately 0.004 ROC-AUC, and both models should practically be considered comparable given the uncertainty associated with the small sample.

The choice of Random Forest was therefore made as a pragmatic modeling decision rather than a claim that it represents a substantially superior predictive model.

An additional advantage of Random Forest for this project is that its nonlinear structure provides a useful setting for demonstrating SHAP-based interpretability.

11. Model Interpretability with SHAP

11.1 Motivation

Because Random Forest does not provide directly interpretable coefficients, SHAP (SHapley Additive exPlanations) was used to examine how individual features contributed to model predictions.

The goal of this analysis was not to improve predictive performance, but to understand the behavior of the selected model and determine whether its learned patterns were clinically plausible.

SHAP values were calculated using TreeExplainer for the positive class, representing the predicted probability of a prolonged ICU stay.

The SHAP implementation was validated by confirming that the base value plus the sum of feature-level SHAP contributions reconstructed the model's predicted probability for every observation.

11.2 Global Feature Importance

The eight most important features based on mean absolute SHAP value were:

Rank	Feature	Mean SHAP
1	Medication count	0.0441
2	Heart failure	0.0293
3	Age at admission	0.0261
4	Medicare insurance	0.0261
5	First creatinine	0.0234
6	Hypertension	0.0218
7	CKD	0.0201
8	First WBC	0.0186

Medication count was the most influential feature according to the model. Three chronic comorbidity indicators—heart failure, hypertension, and CKD—also ranked among the seven most influential features.

This pattern is clinically plausible because chronic cardiovascular and renal conditions may reflect greater baseline disease burden and increased complexity of ICU management. However, these observations represent associations learned by the model and should not be interpreted as causal effects.

Age and first-24-hour creatinine were also among the most influential predictors. In particular, creatinine demonstrated a relatively clear monotonic relationship with SHAP contribution: higher creatinine values generally increased the predicted probability of prolonged ICU stay.

11.3 Medication Count and Nonlinear Effects

An important finding was that medication count did not have a simple monotonic relationship with predicted risk.

SHAP dependence analysis showed that the contribution of medication count generally increased from lower values toward the middle range, but then decreased substantially among patients with the highest medication counts.

Therefore, the model did not learn a simple relationship of:

more medications → longer ICU stay.

Instead, the relationship was nonlinear.

This finding illustrates the value of SHAP beyond a simple feature-importance ranking. It demonstrates that a feature may be highly influential while still having a complex relationship with the predicted outcome.

Several explanations are possible, including clinical heterogeneity or sparse observations in the highest medication-count range. Given the limited sample size, the analysis does not provide enough evidence to distinguish between these explanations.

11.4 Medicare Insurance as a Potential Proxy Variable

`insurance_Medicare` ranked almost identically to age in terms of mean absolute SHAP value.

This result should not be interpreted as evidence that Medicare insurance itself causes prolonged ICU stays.

A more plausible explanation is that Medicare status captures information related to age, since Medicare eligibility is strongly associated with older age. The model may therefore be using insurance status as a proxy for demographic and health characteristics.

This observation highlights an important fairness consideration in clinical machine learning: variables that appear predictive may partially encode other demographic characteristics rather than representing independent clinical mechanisms.

11.5 Local Explanations

Three held-out test cases were examined using SHAP waterfall plots.

Case 1: Correct Positive Prediction

The first patient had a predicted probability of 0.566 and an actual prolonged-stay label of 1.

The prediction was pushed toward prolonged stay by factors including:

- medication count
- Medicare insurance
- age
- hypertension

These positive contributions were partially offset by the absence of heart failure and CKD and a relatively normal creatinine level.

This case illustrates how the model combined several moderate risk signals rather than relying on a single dominant feature.

Case 2: Correct Negative Prediction

The second patient had a predicted probability of 0.178 and an actual label of 0.

Interestingly, the patient had a relatively high medication count of 31. However, SHAP assigned a strong negative contribution to medication count in this range.

The absence of comorbidities and normal hemoglobin further pushed the model toward a non-prolonged prediction.

This case provides a concrete example of the nonlinear medication-count relationship observed in the global dependence analysis.

Case 3: Incorrect Positive Prediction

The third patient was incorrectly classified as prolonged, with a predicted probability of 0.546 despite an actual non-prolonged outcome.

The patient was older, had Medicare insurance, heart failure, and hypertension. These features strongly pushed the prediction toward prolonged stay.

However, the actual ICU course was short.

This error is particularly informative because it demonstrates that clinically plausible risk factors do not deterministically determine ICU length of stay. The model captured statistical associations, but these associations were insufficient to correctly classify every individual patient.

12. Limitations

Several limitations must be considered when interpreting the results.

12.1 Small Sample Size

The final dataset contained only 117 ICU stays and 31 prolonged-stay cases. The test set contained only 24 observations and six positive cases.

As a result, performance metrics can change substantially with a small number of different predictions. The cross-validation mean and standard deviation therefore provide more useful context than any single test-set metric.

12.2 MIMIC-IV Demo Dataset

This project used the MIMIC-IV Clinical Database Demo rather than the complete MIMIC-IV database.

The purpose of the project was therefore to demonstrate a reproducible data engineering and machine learning workflow rather than produce a clinically validated prediction model.

12.3 Limited Feature Set

Only 17 predictive features from six source tables were included.

Important ICU information such as detailed vital signs, fluid balance, nursing observations, procedures, and standardized severity scores was intentionally excluded from this first-stage project.

In particular, the dataset does not include SOFA, APACHE, or SAPS severity scores.

12.4 Weak Predictive Performance

The three models achieved ROC-AUC values ranging from approximately 0.48 to 0.59.

The relatively weak performance is consistent with the small sample size and limited feature set. It also reflects the weak individual feature-target relationships identified during EDA.

Importantly, the project does not claim that these models are clinically useful.

12.5 Fixed Binary Outcome

ICU length of stay was converted into a binary outcome using a five-day threshold.

This transformation simplifies the prediction task but discards information about the exact duration of ICU stay.

A future version of the project could instead treat length of stay as a continuous or time-to-event outcome.

13. Future Work

The project can be extended in several directions if access to the complete MIMIC-IV database is available.

First, additional first-24-hour clinical information could be incorporated, including detailed vital signs, fluid balance, and additional laboratory measurements.

Second, standardized severity scores such as SOFA, APACHE II/IV, and SAPS could be constructed and evaluated as baseline clinical predictors.

Third, a standardized comorbidity index such as the Charlson or Elixhauser Comorbidity Index could replace the small set of manually selected chronic disease indicators.

Fourth, rather than predicting a binary prolonged-stay label, future work could model ICU length of stay directly using survival analysis or regression approaches. Survival analysis would preserve more information and could naturally account for time-to-discharge.

Finally, with a substantially larger cohort, more advanced temporal models could be evaluated using sequential ICU observations rather than a static first-24-hour feature snapshot.

14. Conclusion

This project developed an end-to-end machine learning pipeline for early prediction of prolonged ICU length of stay using the MIMIC-IV Clinical Database Demo.

The project began with raw multi-table EHR data and used DuckDB-based SQL processing to construct a stay-level analytical dataset while enforcing a strict first-24-hour information boundary to prevent temporal leakage.

Following EDA and feature engineering, Logistic Regression, Random Forest, and XGBoost were compared using stratified train/test evaluation and five-fold cross-validation. Predictive performance was modest, with all models remaining relatively close to chance-level discrimination. Given the small sample

size and limited feature set, these results were treated as methodological observations rather than evidence of clinical performance.

Random Forest was selected as the final model based on its highest mean cross-validation ROC-AUC, although its advantage over XGBoost was minimal. SHAP analysis was subsequently used to interpret the model and identify the features most strongly associated with its predictions.

Medication count, heart failure, age, Medicare insurance, creatinine, hypertension, and CKD emerged as important model features. The analysis also identified nonlinear behavior in medication count and highlighted Medicare insurance as a potential proxy for age rather than an independent clinical driver.

Overall, the primary contribution of this project is not a high-performing clinical prediction model. Instead, it demonstrates the ability to transform messy relational EHR data into a reproducible, leakage-conscious machine learning pipeline, evaluate models appropriately under severe sample-size constraints, and critically interpret model behavior without overstating weak or potentially confounded findings.